Applied Machine Learning and Predictive Modelling 1

Evaluation information and hints for the group work, HS26

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Admin

In order to obtain the credits for the course "Applied Machine Learning and Predictive Modelling I" students must deliver a group work and take part to the written exam. The written exam will take place during the exam session at the end of the semester (the exact exam date is not known yet).

Repeating Students

Repeating students **must write to the module coordinator** (Nisia Trisconi), whether they will do the group work, take the exam or both. This information must also be communicated to Tara Meulenbeld. Deadline is the deadline for the repetition modules ("Anmeldefrist der Repetitionsmodule").

Written exam

- It is planned to take place in Luzern (physical presence).
- It lasts for 60 minutes.
- It will account for 60% of the final grade.
- Examination language is English.
- It does not imply the use of computers **nor** calculators. You may have to complete some **R**-codes or to write some pseudo-code.
- It is "closed book".
- You are allowed to take a summary with you.
  - The summary must be hand-written on a A4 sheet (both sides allowed).
  - NB: it must be written by yourself! Versions made on a digital device are allowed, but **must be printed out**.
  - Summaries that are too similar/identical will be confiscated and the exam grade penalised by 0.5 points.
  - Indeed, the aim of the summary is to have you learn while producing it. Simply taking a good summary from someone else does not bring much to the learning process.

Group work

Administrative information

- It will account for 40% of the final grade.

- It must be written in English.
- It represents a complete analysis of a data set where you use the machine learning methods seen in class.
- In particular, you must use:
  - a linear model
  - a Generalised Additive Model
  - a Generalised linear model with family set to Poisson **OR** Binomial (one is compulsory, the other is optional)
  - a Neural Network
  - a Support Vector Machine
- You also have to include one paragraph about
  - how you used generative AI in redacting the group work (code-related questions, generate text, explain concepts, …)
  - what was easy/hard/impossible to do with generative AI
  - what you had to pay attention to/be critical about when using the results obtained through the use of generative AI
- It must be uploaded on ILIAS in a dedicated delivery folder.
- **the deadline to deliver the group work is on Friday two weeks (for FS) and resp. three weeks (for HS) after the semester end at 5pm (i.e., in HS26 the deadline is on January 8th 2027, at 5 PM).**
- NB: the deadline is hard. Late submissions will be penalised by **one** point (i.e. maximum grade achievable is then 5).
- Email submission are not accepted. The work must be submitted to ILIAS (even in case of late submissions).
- It is done by groups of **three** students.
- The group composition (names) are communicated to us by entering them in the corresponding survey on ILIAS. Deadline for communicating the group composition and choosing the data set is week 4.
- NB: **it is not possible to have mixed groups composed by repeating and first attempt students**.
- NB: if you fail to submit together as a group, everyone in the group will be penalised.

Data set choice

- The group work represents a complete analysis of a real data set.
- The choice of the data set to be analysed is free. Nevertheless, the data set must fulfill the following requirements:
  - Moderate size (N = [10^3\$10^3\$, 10^5], 10-20 predictors).
  - Please do not use data sets with more than 10^5 observations. If you find a data set that has more than this amount of observations, you can consider a subset of it.

  - Real data.
  - Must contain variables of the following type
    * continuous (i.e., value can be any number)
    * categorical (e.g., values of this variable are "brown", "orange", "blue")
      · At least one categorical variable must have more than two levels.
    * count (i.e., a variable that is positive and has integer values: 0, 1, 2, 3, 4, …) **OR** binary (i.e., something that has only two levels, 0 and 1 or "good" and "bad").
      · one is compulsory, the other is optional.

- Kaggle is just one of the many sources of data sets. You may want to consider websites like
  - https://zenodo.org
  - www.opendata.swiss
  - https://opendata.swisscom.com
  - https://datasetsearch.research.google.com/
  - http://puenktlichkeit.ch/
  - https://data.world/uci/
  - www.mldata.io
  - https://www.pxweb.bfs.admin.ch/
  - https://www.ostluft.ch/
  - http://www.agrometeo.ch/de/meteorology/datas
  - https://data.stadt-zuerich.ch/
  - https://data.gov.sg/dataset (just to mention a few examples…).
- **R** itself comes with a few hundreds data sets (type "data()" to see the list), the vast majority of **R** add-on packages also comes with data sets.
- Note also that if you find a cool data set that does not fulfill all the requirements, you may complement it with other data sets (e.g., by adding coordinates, weather data or similar things).
- Note that the data set used by the lecturers cannot be reused for your work.
- You can also use data from your employer. If you do so, please make sure that you are allowed to share this data with us and your colleagues!

Deliverables

- The core deliverable piece of work is a pdf/html file created with Rmarkdown.
- Note, however, that we want you to deliver more than just the pdf/html file.
- In particular, you are expected to upload a zipped file that contains:
  - The raw data.

  - A pdf/html file that documents the data preparation, graphical analysis, modelling and comparison of the models.
  - The corresponding **R**-code (and .Rmd file).
  - The files must be structured in folders properly documented via a "ReadMe" file.

- The work must be uploaded on ILIAS as a .zip file with the following properties:
  - name your main folder (and corresponding .zip file) as `FamilyNameStudent1_FamilyNameStudent2...`

- You should also add the matriculation number of one of the students
  - So, the naming should look like `FamilyNameStudent1_FamilyNameStudent2...MatriculationNumber.zip` (e.g. `Trisconi_Meister_Tanadini_010923.zip`).

- Make sure that the work is uploaded by the deadline.
  NB: you can submit a pre-final version if you like. If you do so, make sure that you clearly name your files according to the following scheme: "Trisconi_Meister_Tanadini_PreFinal.zip".
  - **we will only grade the last version you uploaded on ILIAS.**

Rmarkdown and collaboration

- **Do compile often to avoid last minute issues with files that do not compile properly.** Every year there are students who don't hand-in the required html/pdf file because they didn't compile their files often enough and then had problems at the very end of the project.
- Do find an appropriate and efficient way to collaborate with the other group members (e.g., Git, GitHub, Dropbox, …).

Analysis structure and content

- The report must contain an analysis of the chosen data set with all methods (that is, the models) seen in class (however, one between GLM Poisson and GLM Binomial is optional). Remember to also include a graphical analysis!
- For **each** model:
  - you should **interpret the coefficients** that you get from the model (this is also a great check to see whether you did something that makes sense or not and is a great way to prepare yourself for the exam).

- The total length of the document **cannot exceed 30 pages**. If you are submitting an html document, check its length!

- Cross validation or other methods for model comparison must be used on one single method (e.g., you use Cross Validation to compare 2-3 SVM models).
- Students are free to choose a measure of fit that they find more appropriate.
- In case students cannot find an appropriate measure of fit, they can use the Root Mean Squared Error (RMSE).
- **Do** comment on the results of all phases. **This means interpreting the coefficients and translating these results in a way that a reader can understand what the model is telling you. Simply looking at p-values is not enough.**
- Also write a short "Conclusions" paragraph.

Personal contribution

- All three group members must contribute in "similar/same" way to the group work.
- Each of the three group members must take the lead on a a given model. This must be indicated in the pdf file (e.g., write, just after the title of the linear model section "Nisia Trisconi took the lead on this method").
- Note that this does not mean that the person will need to work on that chapter on their own. On the contrary, as the name implies, the group work is supposed to be a team work.
- As there are more than three methods to use in the group work, only three of them need to have a clear lead from one of you.
- You should fill the "team agreement" and upload it into the dedicated folder on ILIAS.

Assessment criteria for the group work

- We have intentionally chosen not to provide a fixed set of assessment criteria, as we do not want to limit your focus to a small number of predefined aspects or encourage you to optimise specifically for them.
- Instead, we encourage you to produce a well-rounded and thoughtful analysis overall.
- **Your final aim should be satisfying a potential client**.
- So, act as you were producing this document for a client.
- The main point is that, given a data set to analyse, there is no "best solution". There are many different ways to produce an excellent analysis. Some analyses may focus more on model interpretations, another analysis may focus on visual aspects of the graphs, yet another analysis may focus on a sophisticated pdf/html document… In other words, there are many ways on how you could get a 6 for the group work.
- Be creative!

Hints for the group work

Compile often

- Avoid compiling your .Rmd file the day before the deadline.
- If you don't compile often, the .Rmd file is very likely to not compile properly. You can compile every time you add a few lines of code.
- If you compile your .Rmd file after adding a few lines of code and the Rmd does not compile, you know where to search for the mistake.
- If you changed very many things in your code and you get troubles in compiling, then it is much more difficult to understand what new line of code is creating the issue.
- If you get compilation troubles and can't find the mistake, you can start commenting out portions of your code. If then the file compiles properly… you know that the line creating issues is among those that you just commented out.
- Finally, note that compiling documents may take some time… especially when complex models are fitted.
- In these cases you can use the argument option `cache = TRUE` such that a chunk is newly evaluated only if it was modified since the last compilation. If the chunk remained unchanged, then the old results are used.
- Alternatively, you can save fitted models as .RDS files (see the `saveRDS()` function) such that the model does not need to be re-fitted at each compilation. You can then load the fitted model with the `readRDS()` function.
- Google for examples to best understand the hints about caching and using .RDS files.

Write a nice story

- This analysis must ba a story that you want to tell to a client.
- There are many ways to tell a nice story.
- Be creative.
- Make use of your strengths and interests:
  - you are interested in parameter interpretation or bootstrap confidence intervals? ⇒ then put emphasis on this.
  - you are very good at creating fancy html documents? ⇒ then spend time on this.
  - you want to deepen your knowledge of SVMs? ⇒ then expand on that topic and add things we haven't seen in class.
- Remember, nevertheless, that the overarching goal is to "produce an analysis of your data set and use all methods".

Use methods in a sensible way

- Don't forget that some methods are suited to almost all situations, others are not.
- Indeed, you can fit a SVM to count data as well as to binary data.
- However, it makes no sense to fit a plain linear model to count or binary data. You should use GLMs instead (with the appropriate family).
- So, fit an LM only to a response variable which is continuous.
- If all the numerical variables in your data set are count data
  - it means that you didn't read the rules on how to choose the data set.
  - nevertheless, you can still transform the count data with the square-root function and fit a linear model (`lm(sqrt(some.counts) ~ ...)`).

Compare methods in a fair way

- The aim of your analysis strongly depends on the story that you created.
- In particular, if you tell a story where a predictive model is needed (which is often the case here) then prediction is the performance to be assessed.
- Note that you may want to fit models to make good predictions, but also to better "understand the data". In other words, prediction is often only a part of the job.
- However, when comparing predictive performance, remember to compare models in a fair way.
- A very common mistake is to fit a very simple linear model, with zero non-linearities, with no interactions, and then to compare it to a complex non-parametric method such as NN or SVN.
- Most likely the very simple model will have a predictive performance that is well below the other methods.
- So, if you want to compare LM with other techniques in a fair way use complex models that contain interactions, non-linearities and even better both of them.
- Note that you may still want to fit a simple linear model that is easy to interpret alongside more complex ones that are more suited for prediction.

Log-transform "amounts"

- The vast majority of the data sets analysed for the group work contain continuous variables that are amounts (see course material if you forgot what amounts are).
- Typical examples of amounts seen in the group works are:
  - weight
  - income
  - costs

- Most of the time these response variables are modelled with a linear model, which is a correct choice for continuous variables.
- **NOTE, however, that "amounts" need virtually always to be log-transformed before being analysed** (so, graphically check if this is needed!)
- If a log-transformation is needed, you should fit a model like `lm(log(costs) ~ ...)`.

Length of the document

- As mentioned above, brevity is important. You are not allowed to hand-in anything longer than 30 pages.
- If you are not producing a pdf, but an html file you can
  - try to render the html as a pdf to see whether it exceeds 30 pages.
  - simply look at the print preview and see how many pages it is.
  - note that working with an html file as an output is fully ok!
  - if you produce an html file, submit it together with a "check" that it does not exceed the 30 pages limit.
  - we have absolutely no preference between pdf and html.
- Make use of simple tricks to shorten your document.
  - e.g., omit messages such as those created when loading add-on packages (see the "message" chunk option).
  - if the output of a model is too long… omit it or cut it (e.g., with the `capture.output()` function).
- In general, it is bad practice to omit code though.
- Indeed, without code, it is difficult to understand what has been done.
- In html documents you can add a button to show/hide code.

There are plenty of data sets

- Avoid working on the same data sets as other groups!
- There is plenty of data sets freely available, so really avoid this.
- Kaggle is a popular source of data sets for this group work. Note, however, that often kaggle data sets come with existing analyses, which in turn questions your actual contribution and creativity. Using a non-kaggle data set is taken as a sign of engagement and genuine work from your side.

Complementary information and project structure

Please structure your submission by complementing it with the following information:

- A README file to describe what all files in the submitted .zip file are.
- The source of the data set (e.g., a website or a company that provided the data).
- A brief "contextual" description of the project (e.g., where the data comes from and so on).
- Authors.

If you want, you can complement the README file with information about who did what in your group.

Groups of three people

If your group is not composed by three people as required, you should explain why. In case some students left the master, you should state who were the team members at the beginning. Teams that are not composed by three students who don't provide an explanation will be penalised.

Write conclusions

- Do write conclusions.
- This implies commenting and interpreting the results of your analyses as well as writing a short paragraph with conclusions at the end of your analysis.
- Brevity is essential here.
- Don't write very long interpretations for any single plot, just what is needed.
- Keep in mind that a client will be reading your report. So, you want to be informative, but not boring.

FAQs

- Do we need to use all methods in our analysis? ⇒ Yes, do use all methods seen in class (LM, GLMs, …). However, one between GLM Poisson and GLM Binomial is optional. You can choose to use both if you like, but at least one of them must be used.
- We realised our data set is not so well-suited as we believed, can we change it even though the deadline for choosing a data set is overdue? ⇒ In principle you should avoid changing data set during the group work. No data set is perfect and it can well happen that none of the methods finds any structure in the data (welcome to real data science).
- Why is there no example of such an analysis, we struggle in understanding what is required! ⇒ There is no example because then all students would stick to it. We want you to be creative and come up with interesting stories. You can use the Labs as inspiration though.
- Our data set is very large and models take ages to fit? ⇒ You can work on a sub-sample.
- In your labs you fit linear models with one predictor only, should we do that too? ⇒ NO, NO, NO! Please do not fit linear models with one predictor only. This is only done for didactic purposes. In reality, you never fit a linear model with one predictor only. **Do fit models were all selected predictors are considered**.
- The residuals of our linear model don't seems to be normally distributed and their variances do not seem to be stable, shall we use another dataset? ⇒ No, residual diagnostics are not the main focus in this course. So, there is no need to change your dataset. **Note, however, that in the vast majority of the cases you simply need to log-transform the response variable when fitting a linear model and the model assumptions are better fulfilled.** In particular, if the response variable are "amounts" (continuous variables that can only take positive values such as costs, weight, length…) then you must log-transform them!
- Shall we add a regression line on the exploratory graphs? ⇒ No, please always add smoothers to exploratory graphs (i.e., graphs where the response variable is plotted against a predictor).
- We found a very interesting data set, but it as too many predictors (e.g., 150), shall we find another data set? ⇒ No, you can work on a sub-sample of all the predictors. Just 10-20 predictors that you find most interesting.
- We found a very interesting data set, but it as too many observations (e.g., 350'000), shall we find another data set? ⇒ No, you can work on a sub-sample of all the observations. Just take a **random** sub-sample of 1-10k observations.
- Can we use the data set used for the R-bootcamp? ⇒ Yes, you can. Make sure that the overlapping between the two reports is minimal though.